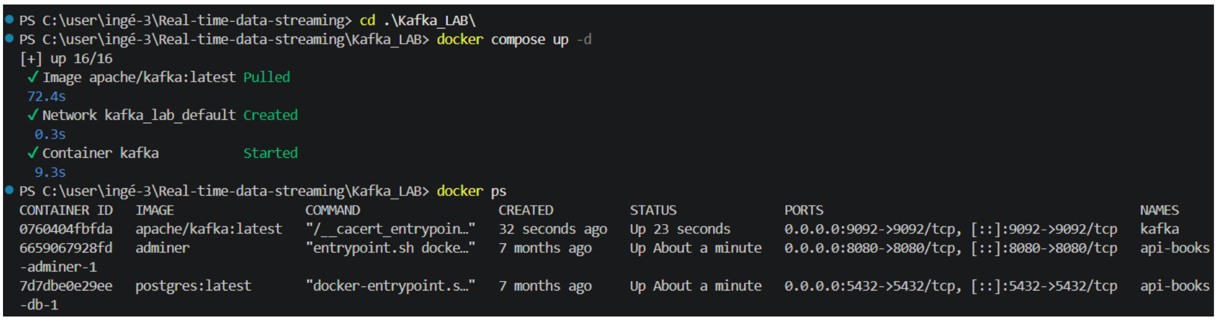
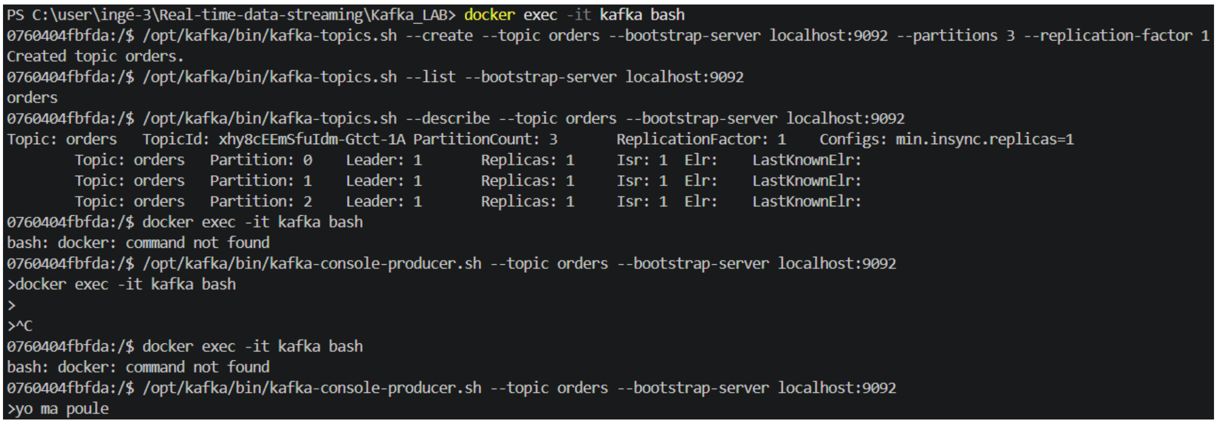
Léonie Schmit 22/10/2026

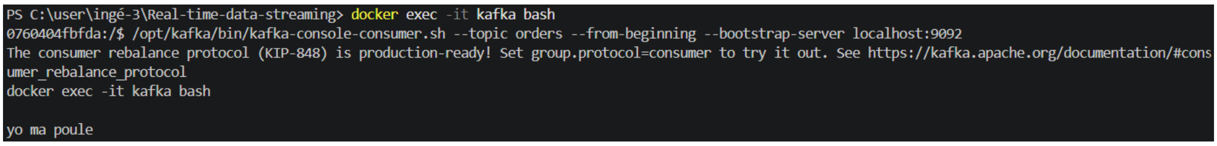
Tp 1 : Kafka



```
PS C:\user\ingé-3\Real-time-data-streaming> cd .\Kafka_LAB\
PS C:\user\ingé-3\Real-time-data-streaming\Kafka_LAB> docker compose up -d
[+] up 16/16
 ✓ Image apache/kafka:latest Pulled
 72.4s
 ✓ Network kafka_lab_default Created
  0.3s
 ✓ Container kafka           Started
  9.3s
PS C:\user\ingé-3\Real-time-data-streaming\Kafka_LAB> docker ps
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS              PORTS                                         NAMES
0760404fbfda   apache/kafka:latest   "/__cacert_entrypoin…"   32 seconds ago   Up 23 seconds       0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp   kafka
6659067928fd   adminer               "entrypoint.sh docke…"   7 months ago     Up About a minute   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp   api-books
-adminer-1
7d7dbe0e29ee   postgres:latest       "docker-entrypoint.s…"   7 months ago     Up About a minute   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp   api-books
-db-1
```

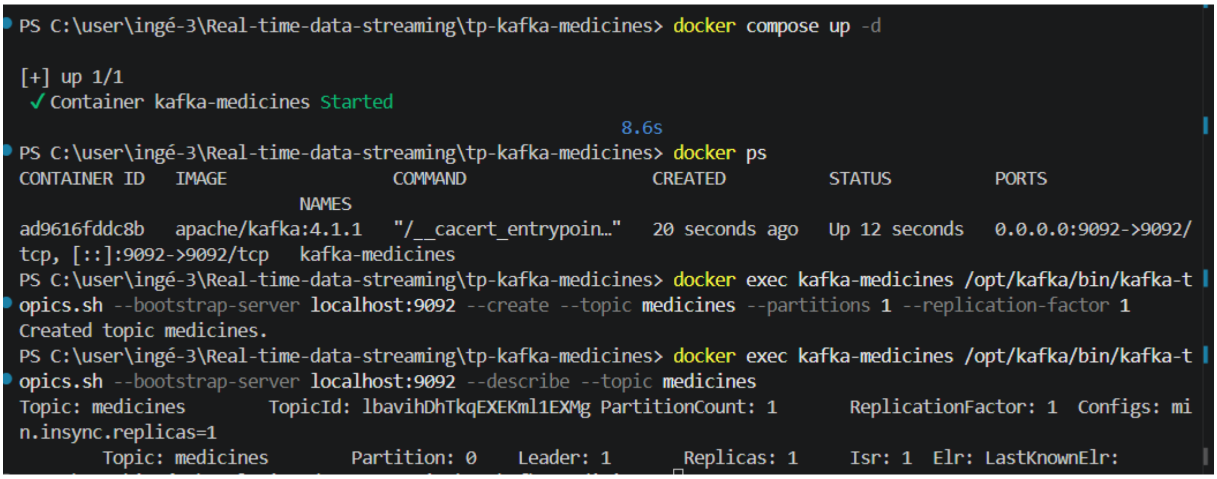


```
PS C:\user\ingé-3\Real-time-data-streaming\Kafka_LAB> docker exec -it kafka bash
0760404fbfda:/$ /opt/kafka/bin/kafka-topics.sh --create --topic orders --bootstrap-server localhost:9092 --partitions 3 --replication-factor 1
Created topic orders.
0760404fbfda:/$ /opt/kafka/bin/kafka-topics.sh --list --bootstrap-server localhost:9092
orders
0760404fbfda:/$ /opt/kafka/bin/kafka-topics.sh --describe --topic orders --bootstrap-server localhost:9092
Topic: orders   TopicId: xhy8cEEmSfuIdm-Gtct-1A PartitionCount: 3       ReplicationFactor: 1    Configs: min.insync.replicas=1
        Topic: orders   Partition: 0    Leader: 1       Replicas: 1     Isr: 1  Elr:    LastKnownElr:
        Topic: orders   Partition: 1    Leader: 1       Replicas: 1     Isr: 1  Elr:    LastKnownElr:
        Topic: orders   Partition: 2    Leader: 1       Replicas: 1     Isr: 1  Elr:    LastKnownElr:
0760404fbfda:/$ docker exec -it kafka bash
bash: docker: command not found
0760404fbfda:/$ /opt/kafka/bin/kafka-console-producer.sh --topic orders --bootstrap-server localhost:9092
>docker exec -it kafka bash
>
>^C
0760404fbfda:/$ docker exec -it kafka bash
bash: docker: command not found
0760404fbfda:/$ /opt/kafka/bin/kafka-console-producer.sh --topic orders --bootstrap-server localhost:9092
>yo ma poule
```

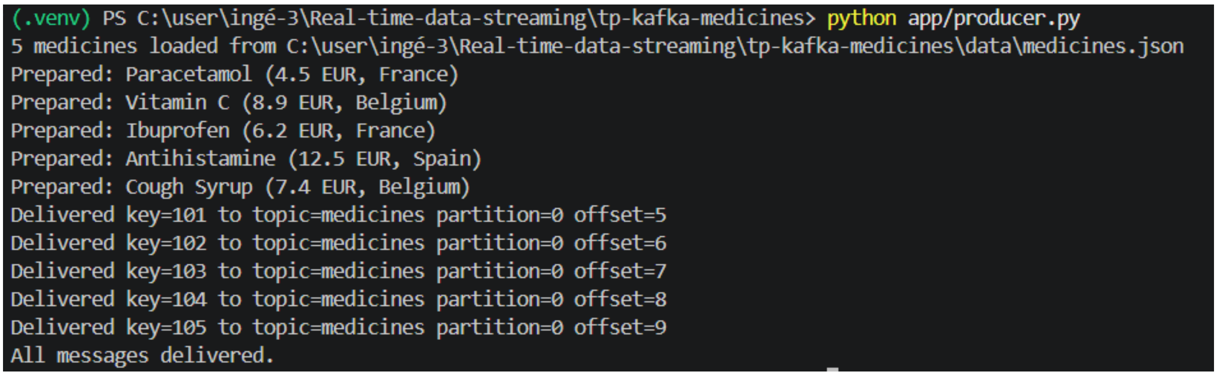


```
PS C:\user\ingé-3\Real-time-data-streaming> docker exec -it kafka bash
0760404fbfda:/$ /opt/kafka/bin/kafka-console-consumer.sh --topic orders --from-beginning --bootstrap-server localhost:9092
The consumer rebalance protocol (KIP-848) is production-ready! Set group.protocol=consumer to try it out. See https://kafka.apache.org/documentation/#cons
umer_rebalance_protocol
docker exec -it kafka bash

yo ma poule
```

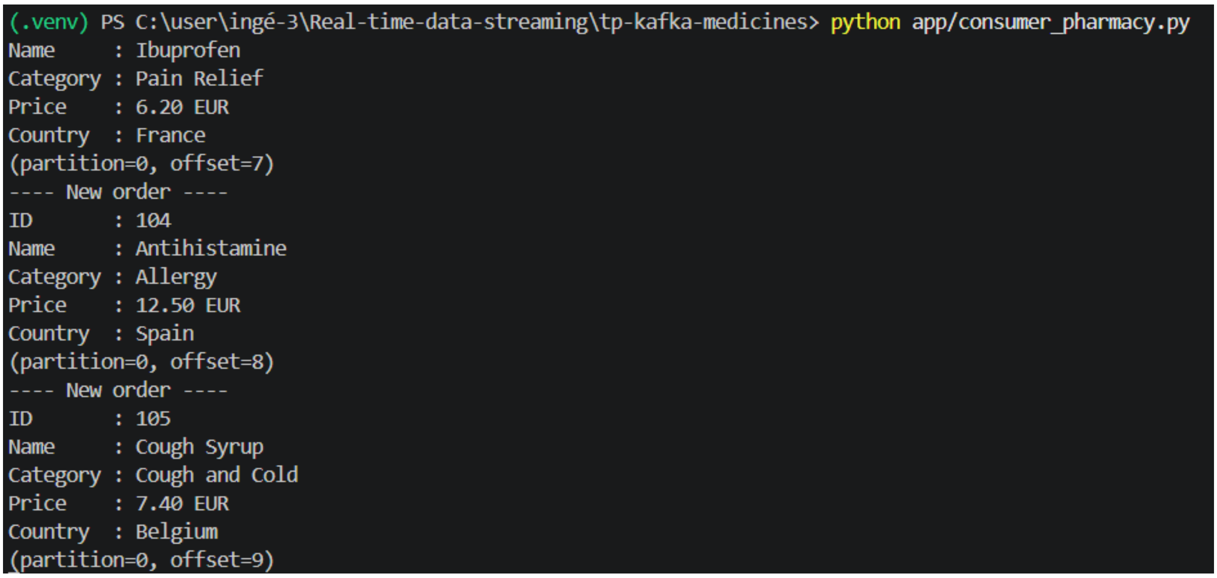
Tp 2 :



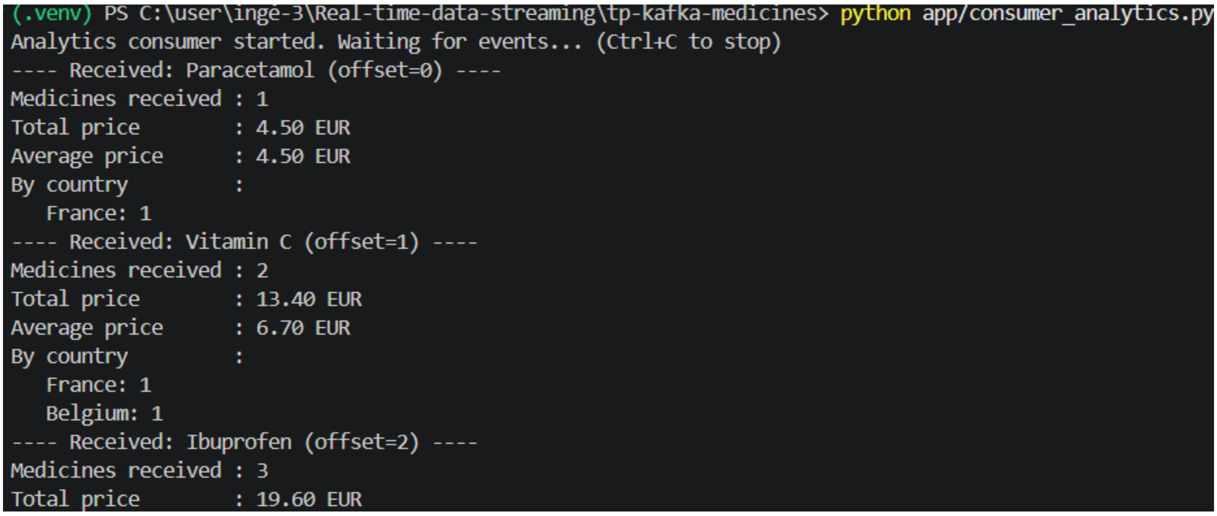
```
PS C:\user\ingé-3\Real-time-data-streaming\tp-kafka-medicines> docker compose up -d

 [+] up 1/1
  ✓ Container kafka-medicines Started
                                                            8.6s
PS C:\user\ingé-3\Real-time-data-streaming\tp-kafka-medicines> docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS          PORTS
                           NAMES
ad9616fddc8b   apache/kafka:4.1.1   "/__cacert_entrypoin…"   20 seconds ago   Up 12 seconds   0.0.0.0:9092->9092/
tcp, [::]:9092->9092/tcp   kafka-medicines
PS C:\user\ingé-3\Real-time-data-streaming\tp-kafka-medicines> docker exec kafka-medicines /opt/kafka/bin/kafka-t
opics.sh --bootstrap-server localhost:9092 --create --topic medicines --partitions 1 --replication-factor 1
Created topic medicines.
PS C:\user\ingé-3\Real-time-data-streaming\tp-kafka-medicines> docker exec kafka-medicines /opt/kafka/bin/kafka-t
opics.sh --bootstrap-server localhost:9092 --describe --topic medicines
Topic: medicines        TopicId: lbavihDhTkqEXEKml1EXMg PartitionCount: 1       ReplicationFactor: 1  Configs: mi
n.insync.replicas=1
        Topic: medicines        Partition: 0    Leader: 1       Replicas: 1     Isr: 1  Elr: LastKnownElr:
```



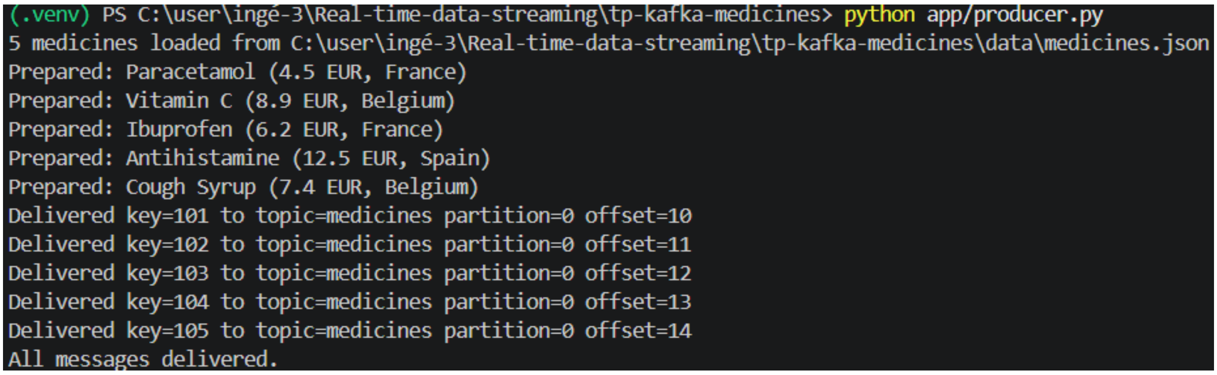
```
(.venv) PS C:\user\ingé-3\Real-time-data-streaming\tp-kafka-medicines> python app/producer.py
5 medicines loaded from C:\user\ingé-3\Real-time-data-streaming\tp-kafka-medicines\data\medicines.json
Prepared: Paracetamol (4.5 EUR, France)
Prepared: Vitamin C (8.9 EUR, Belgium)
Prepared: Ibuprofen (6.2 EUR, France)
Prepared: Antihistamine (12.5 EUR, Spain)
Prepared: Cough Syrup (7.4 EUR, Belgium)
Delivered key=101 to topic=medicines partition=0 offset=5
Delivered key=102 to topic=medicines partition=0 offset=6
Delivered key=103 to topic=medicines partition=0 offset=7
Delivered key=104 to topic=medicines partition=0 offset=8
Delivered key=105 to topic=medicines partition=0 offset=9
All messages delivered.
```



```
(.venv) PS C:\user\ingé-3\Real-time-data-streaming\tp-kafka-medicines> python app/consumer_pharmacy.py
Name     : Ibuprofen
Category : Pain Relief
Price    : 6.20 EUR
Country  : France
(partition=0, offset=7)
---- New order ----
ID       : 104
Name     : Antihistamine
Category : Allergy
Price    : 12.50 EUR
Country  : Spain
(partition=0, offset=8)
---- New order ----
ID       : 105
Name     : Cough Syrup
Category : Cough and Cold
Price    : 7.40 EUR
Country  : Belgium
(partition=0, offset=9)
```



```
(.venv) PS C:\user\ingé-3\Real-time-data-streaming\tp-kafka-medicines> python app/consumer_analytics.py
Analytics consumer started. Waiting for events... (Ctrl+C to stop)
---- Received: Paracetamol (offset=0) ----
Medicines received : 1
Total price        : 4.50 EUR
Average price      : 4.50 EUR
By country         :
   France: 1
---- Received: Vitamin C (offset=1) ----
Medicines received : 2
Total price        : 13.40 EUR
Average price      : 6.70 EUR
By country         :
   France: 1
   Belgium: 1
---- Received: Ibuprofen (offset=2) ----
Medicines received : 3
Total price        : 19.60 EUR
```



```
(.venv) PS C:\user\ingé-3\Real-time-data-streaming\tp-kafka-medicines> python app/producer.py
5 medicines loaded from C:\user\ingé-3\Real-time-data-streaming\tp-kafka-medicines\data\medicines.json
Prepared: Paracetamol (4.5 EUR, France)
Prepared: Vitamin C (8.9 EUR, Belgium)
Prepared: Ibuprofen (6.2 EUR, France)
Prepared: Antihistamine (12.5 EUR, Spain)
Prepared: Cough Syrup (7.4 EUR, Belgium)
Delivered key=101 to topic=medicines partition=0 offset=10
Delivered key=102 to topic=medicines partition=0 offset=11
Delivered key=103 to topic=medicines partition=0 offset=12
Delivered key=104 to topic=medicines partition=0 offset=13
Delivered key=105 to topic=medicines partition=0 offset=14
All messages delivered.
```

Observation :

On startup, both consumers immediately display the messages sent earlier in Part 2. Kafka keeps messages after they're read, and each new group starts from the beginning because of earliest.

When the producer runs, both consumers receive all 5 new messages. They're in different groups (pharmacy-group and analytics-group), and Kafka tracks a separate offset for each group, so each one gets every message. If both were in the same group, they would share the partitions instead. With only 1 partition, one consumer would get everything and the other nothing.

The same message produces two different processing results. The pharmacy consumer displays each order individually. The analytics consumer accumulates statistics; with the Part 2 run plus this one, it will count 10 medicines, because Kafka doesn’t delete old messages when the producer runs again.